Run report: genus-3 build, run 20261013T0320Z

Descriptive record of one run. Interpretation is in the study `README.md` / `REPORT.pdf`.

What ran

- Command: `rustc --edition 2021 -O genus3_build.rs && ./genus3_build 5 1 1 && ./genus3_build 7 3 2 && ./genus3_build 11 1 1`.
- Spec: `PROTOCOL.md`, frozen before the run. Deterministic: no randomness, cases taken in index order.
- Isolation: none; shared host under heavy load. No timing is a result. All recorded values are exact counts.
- Raw records: `results/run-20261013T0320Z.jsonl` (`rec` = case, search).

```
run: 20261013T0320Z
command: rustc --edition 2021 -O genus3_build.rs && ./genus3_build 5 1 1 && ./genus3_build 7 3 2 && ./
genus3_build 11 1 1
commit (base): ecb69d15aeb2eba2326ffa6f6fe4e2737efb05d1
source sha256: a10f0d2d08fa4f82aaf9018f0988efa90d5bdfba3b2a3247763c7abe7b461552
rustc: rustc 1.93.1 (01f6ddf75 2026-02-11) (Homebrew)
host: Darwin 25.6.0 arm64, Apple M4 Pro, 14 cpus
uptime at start: 20:20  up 1 day, 19:59, 11 users, load averages: 82.50 101.15 108.47
args 5 1 1: real 0.21
args 7 3 2: real 0.12
args 11 1 1: real 0.65
uptime at end: 20:20  up 1 day, 19:59, 11 users, load averages: 82.50 101.15 108.47
regenerated after PR review: counters renamed and completed (points_skipped ->
e0_affine_points_nonsquare_uu1 + e0_ramification_points_y0_zero + 2 infinity), pushforward_on_E0 ->
projected_points_on_E0, E0 Weierstrass model added. Original source sha256
111247515fcec6b0967217f6d1e32ba82c2546ea1e04aebbdd96a3de2ff7e849; all previously recorded fields reproduced
identically. Timings above are from the original run.
```

Runs requested and completed

Arm	Requested	Completed	Candidates examined / skipped as linearly dependent
q = 5	1 split, 1 irreducible	2	634 / 620
q = 7	3 split, 2 irreducible	5	2443 / 2394
q = 11	1 split, 1 irreducible	2	14657 / 14630

Recorded values

Unit: number of points (N1, N2, N3, divisor checks) or group elements (orders). Exact counts, no uncertainty. In all 9 cases `descends`, `smooth` and `jac_equals_E0` are true, `N` equals `N_predicted`, and `N3_K_model` equals `N[2]`.

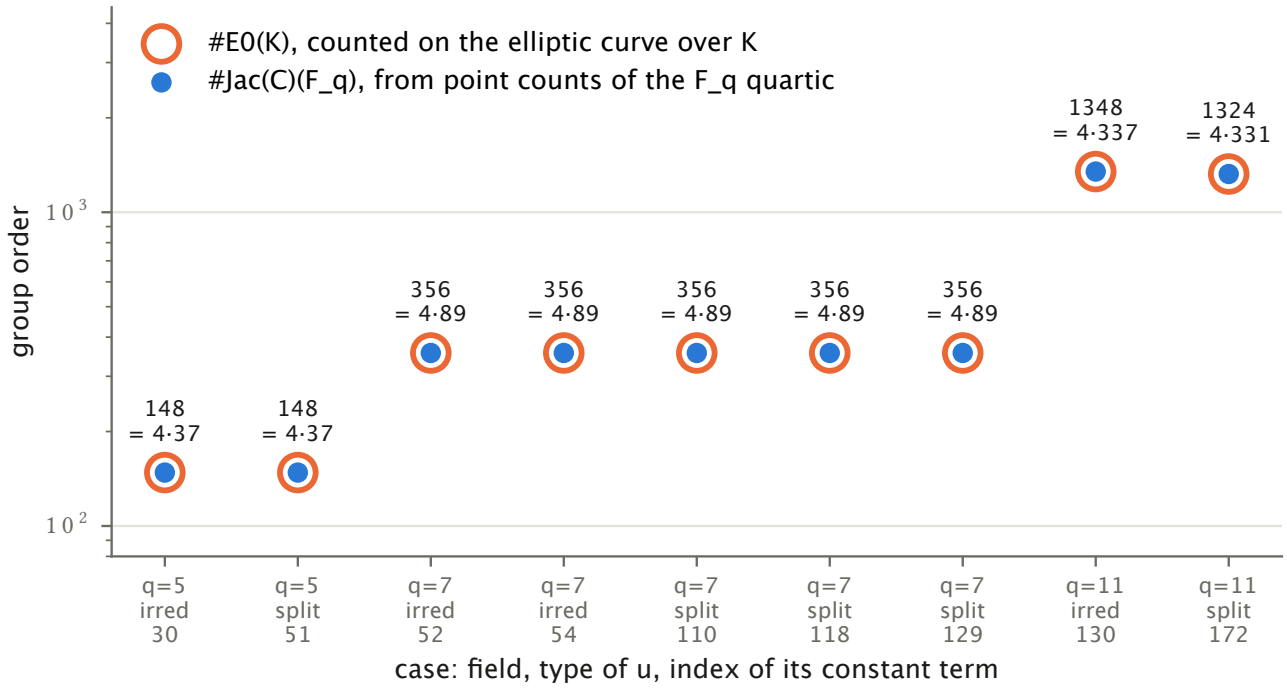


Figure 1: Run 20261013T0320Z, 9 cases. Ring: #E0(K) counted on the elliptic curve. Dot: #Jac(C)(F_q) derived from the three measured point counts of the F_q quartic through the L-polynomial. Log axis.

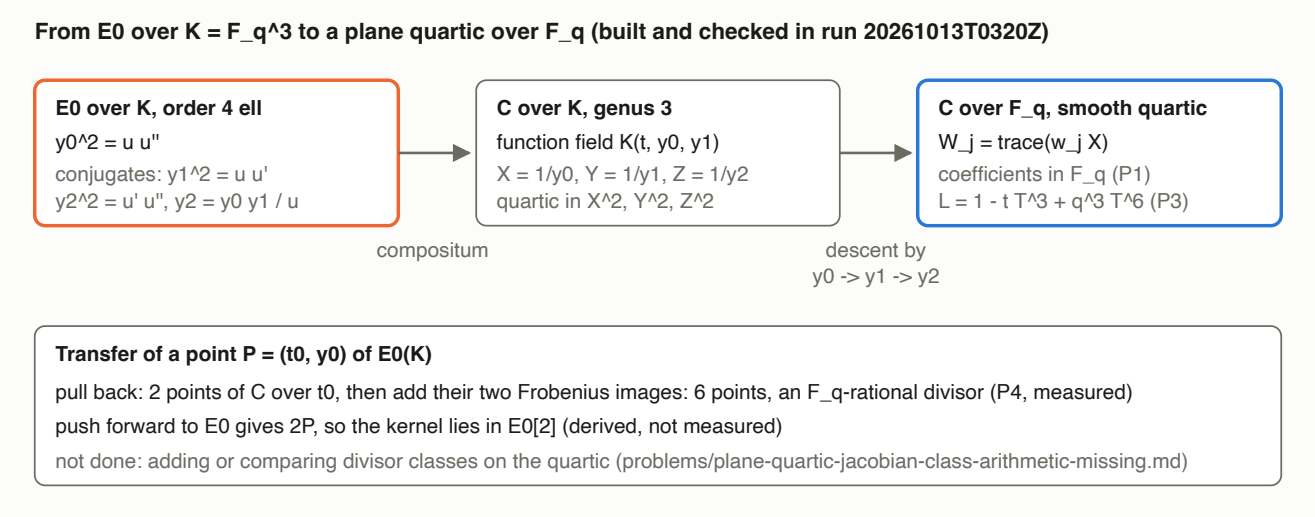


Figure 2: What was built and which steps were measured.

Missing data

- No divisor-class arithmetic on the quartic: homomorphism and injectivity of the transfer were not executed.
- E0 points not tested, exactly accounted (`#E0(K) = tested + nonsquare + ramification + 2`): the 2 points at infinity of the quartic model (always); the points with `y0 = 0` (`e0_ramification_points_y0_zero`: 0 when `u` is irreducible, 4 when `u` splits); and the affine points with `y0 != 0` where `u u'(t0)` is zero or a non-square (`e0_affine_points_nonsquare_uu1`).
- `projected_points_on_E0` only checks that the projected coordinates satisfy the E0 equation; the pushforward of a divisor and `pi0_* phi(P) = [2]P` were not executed.